Concrete

Source: concrete.csv · Target: strength_mpa

Internal regression dataset reliability validation.

1. Regression Review Summary

Regression dataset evidence is available for review. Priorities describe investigation needs, not a score or readiness verdict.

1030 of 1030 targets are finite, with 938 distinct finite values.

Rows	Feature inputs	Review findings	Finite targets
1030	8	3	1030

No validated scoring policy is defined for regression reliability.

Review priorities are investigation needs, not risk bands or a readiness verdict.

> **Relationship coverage: exhaustive within methods**
> No observed relationship means only none verified in the searched scope; unsearched relationships are unknown.

2. Priority Review Actions

Exact repeated records · Moderate review priority

25 rows repeat an earlier complete parsed row (2.43%).

11 groups; 36 participating rows; largest group 4.

Consequence: Repetition changes effective coverage and may matter to later dataset splitting.

Action: Review whether repetition represents valid repeated events, deliberate weighting, or unintended copying. Do not automatically deduplicate.

Population: Name: all complete parsed rows, including the target and nulls; Denominator: 1030. These shared structural observations require context; rarity, repetition, and IQR candidates alone do not prove errors.

Evidence type: review observation · exact_duplicates.5

IQR review candidates · Moderate review priority

Some fields have at least 5% of finite values outside the 1.5-IQR fences.

age_days: 59 of 1030 finite values (5.73%)

Consequence: Extreme observations can influence some analyses; this check does not establish task suitability.

Action: Review distributions, units, and legitimate rare observations. Skew or an IQR flag does not prove a data error.

Population: Name: finite values by affected feature; Denominator: Not set; Detail: Each referenced column retains its own finite denominator; infinity counts use all parsed rows.. These shared structural observations require context; rarity, repetition, and IQR candidates alone do not prove errors.

Evidence type: review observation · numeric_outliers.21

Target variation within repeated inputs · Low review priority

9 feature-identical groups contain varying finite targets.

20 participating rows.

Consequence: Observed variation, support, or repeated-input behavior can affect interpretation of the intended outcome.

Action: Review repeated measurements, stochastic outcomes, omitted context, and recording precision; contradictory labels are not established.

Population: Name: finite targets; Denominator: 1030. Repeated measurements, stochastic outcomes, omitted context, and recording precision can explain variation; contradictory labels are not established.

Evidence type: contextual review · regression_repeated_inputs.30

3. Dataset, Target, and Declared Expectations

Target selected explicitly: **strength_mpa**. Target intent: continuous. Integer-valued observations do not independently establish count semantics.

User-declared expectation	Value
kind	continuous
minimum	Not set
maximum	Not set
nonnegative	Not set
integer required	Not set

User-declared expectation	Value
units	MPa

Units are display context; no unit conversion or semantic validity is inferred.

Column	Role	Parsed dtype	Observed values
cement	feature	float64	1030 / 1030
blast_furnace_slag	feature	float64	1030 / 1030
fly_ash	feature	float64	1030 / 1030
water	feature	float64	1030 / 1030
superplasticizer	feature	float64	1030 / 1030
coarse_aggregate	feature	float64	1030 / 1030
fine_aggregate	feature	float64	1030 / 1030
age_days	feature	int64	1030 / 1030
strength_mpa	target	float64	1030 / 1030

Parsing: Pandas default missing tokens; utf-8. All evidence concerns the parsed interpretation.

4. Target Coverage and Distribution

All-row coverage: 1030 finite, 0 missing, 0 positive infinity, 0 negative infinity.



Distribution population: finite targets, including declared constraint violations. Units: MPa.

Measurement	Finite-target evidence
min	2.3318078
max	82.599225
mean	35.817836
sample std	16.705679
distinct count	938
zero count	0
negative count	0
integer count	0
non integer count	1030
iqr	22.429171
central 90 span	55.845084
bowley skewness	0.042705748
modal share	0.48543689

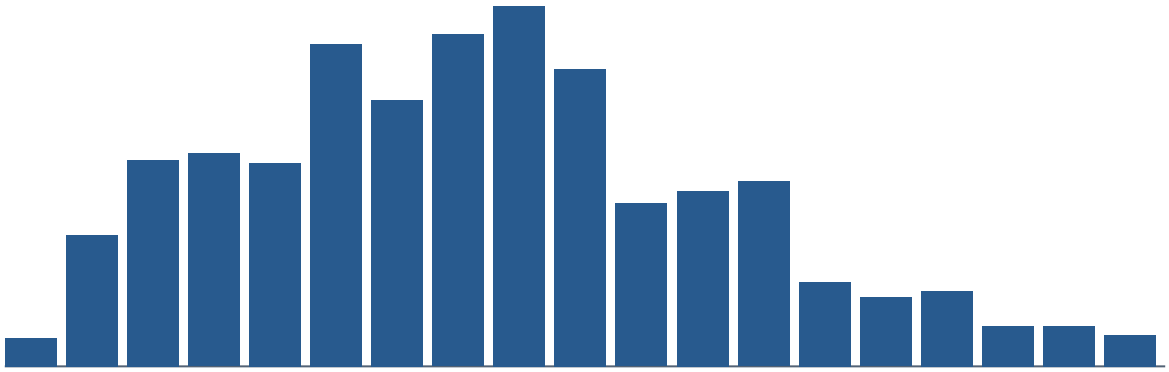
Quantile strip



2.3318078 82.599225

P1 6.889127 · P5 10.959428 · Q1 23.707115 · Median 34.442774 · Q3 46.136287 · P95 66.804512 · P99 77.297154

Finite target distribution



2.3318078 82.599225

freedman diaconis capped 40; counts total 1030 finite observations. Bin counts and interval definitions appear in the appendix.

1.5-IQR candidates: 0 lower, 4 upper, of 1030 finite values. These are review candidates, not established errors.

Declared constraints

0 finite targets violate at least one declared requirement; 1030 satisfy the declared constraints. No observations were removed.

User-declared rule	Violations / finite population
No numerical constraints declared.	

Concentration and numerical representation

Most-common-value share: 0.48543689% of finite targets. Boundary concentration does not establish clipping or saturation.

Boundary	Value	Count / finite share
observed minimum	2.3318078	1 / 0.097087379%
observed maximum	82.599225	1 / 0.097087379%

Numerical representation: completed. Coarsest observed aligned grid: Unavailable. Ordinary integer or fixed-decimal representation is descriptive and generates no rounding action.

5. Target Relationships and Feature Availability

Small or sparse relationships remain qualified technical evidence. No relationship conclusion covers unsearched candidates.

Actions require at least 30 finite overlapping rows, five distinct targets, 80% finite-target coverage, and complete numerical verification. Availability is user-declared or unknown; observed algebra does not prove leakage or actual model use.

Feature	User-declared availability
cement	unknown
blast_furnace_slag	unknown
fly_ash	unknown
water	unknown
superplasticizer	unknown
coarse_aggregate	unknown
fine_aggregate	unknown
age_days	unknown

Distinct relationship groups

No relationship groups were verified in the searched scope. Unsearched candidates remain unknown.

Qualified technical relationship evidence

Observed relationship	Support and finite-target coverage	Interpretation

6. Missingness, Repetition, and Feature Evidence

0 parsed cells missing (0.0%). Rows with missing features: 0 / 1030. Target coverage is reported separately.

cement · 0 missing (0.0%)

blast_furnace_slag · 0 missing (0.0%)

fly_ash · 0 missing (0.0%)

water · 0 missing (0.0%)

superplasticizer · 0 missing (0.0%)

coarse_aggregate · 0 missing (0.0%)

fine_aggregate · 0 missing (0.0%)

age_days · 0 missing (0.0%)

Exact repetition and target variation within repeated inputs require context; automatic deduplication or label-error conclusions are not warranted.

25 repeated rows beyond each first occurrence; 11 exact groups; 36 participating rows.

Repeated-input groups: 19; varying-target groups: 9; varying-target rows: 20.

Example group rows / distinct targets	Target extrema / span
4 / 2	22.897498 to 55.895819; span 32.998321
2 / 2	44.422939 to 51.021224; span 6.5982853
2 / 2	23.345657 to 25.021084; span 1.6754267
2 / 2	38.017707 to 39.300132; span 1.2824254
2 / 2	55.254607 to 56.144031; span 0.88942404

7. Coverage, Methods, Scoring Availability, and Limitations

No validated scoring policy is defined for regression reliability.

Coverage describes these checks, not a percentage of universal reliability.

Numerical shape, rounding, extremes, and repetition can be legitimate. Constraints apply only to declarations. Name cues and algebraic identities do not establish feature availability, causal direction, leakage, or label error. No cleaning, model training, prediction evaluation, or monitoring occurs.

Relationship search: exhaustive within methods. 8 eligible numerical columns; 8 selected for pairs; 44 candidates; 16 fully verified; 0 planned work skips; 0 unfinished candidates after interruption/failure.

Configured relation limits: Max single comparisons: 20000000; Max pair comparisons: 20000000; Max pair features: 32; Max screen rows: 512. Consumed comparisons: Single: 16480; Pair: 14336.

Selected pair columns: cement, blast_furnace_slag, fly_ash, water, superplasticizer, coarse_aggregate, fine_aggregate, age_days.

Support thresholds are internally tested interpretation rules, not externally validated reliability cutoffs.

8. Technical Appendix

Parsing, execution, and coverage disclosure

Parser: pandas.read_csv / pandas 3.0.3. Comma delimiter, inferred header, blank lines skipped. Encoding: utf-8. Missing policy: Pandas default missing tokens. Source values were not rewritten.

Custom missing tokens: None configured; explicit column types: None configured.

Source SHA-256: 938d457c467afb225716d3829152b9931e5951f61f1d1907fd3dcd2bf16e8491. Method: tabular_regression_v1.

Token inspection: 57864 bytes, 200 data records, 1800 comparable cells; stopping reason: record limit. Alignment confirmed: Yes.

Interpretation review warnings are separate from data findings and scoring. Counts describe only inspected source tokens and are never extrapolated to the dataset. No warnings observed does not establish losslessness.

inspection_incomplete
Inspection stopped at record_limit; absence of warnings does not establish losslessness.

Observed occurrences: 0; setting: inference. Confirm this interpretation matches the source meaning; use an explicit parsing override where appropriate.

Operational safeguards

Source/header preflight, bounded row read, post-load frame measurement, cooperative stage deadlines; no hard memory or execution guarantee.

Backend: Pandas in memory; frame: 74292 bytes; elapsed: 0.072896 seconds.

Configured safeguard	Limit
Maximum source size	104857600 bytes
Maximum rows	1000000
Maximum columns	1000
Maximum loaded frame size	536870912 bytes

Configured safeguard	Limit
Cooperative elapsed-time limit	Not set

Check applicability and execution

Check / scope	Applicable	Execution	Reason / exclusions
schema · dataset	applicable	completed	No exclusions
missingness · dataset	applicable	completed	No exclusions
column variation · dataset	applicable	completed	No exclusions
categorical cardinality · dataset	not applicable	not run	No nonnumeric input columns.
feature missingness · dataset	applicable	completed	No exclusions
exact duplicates · dataset	applicable	completed	No exclusions
numeric summary · cement	applicable	completed	No exclusions
numeric outliers · cement	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0
numeric summary · blast_furnace_slag	applicable	completed	No exclusions
numeric outliers · blast_furnace_slag	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0
numeric summary · fly_ash	applicable	completed	No exclusions
numeric outliers · fly_ash	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0
numeric summary · water	applicable	completed	No exclusions
numeric outliers · water	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0
numeric summary · superplasticizer	applicable	completed	No exclusions
numeric outliers · superplasticizer	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0
numeric summary · coarse_aggregate	applicable	completed	No exclusions
numeric outliers · coarse_aggregate	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0
numeric summary · fine_aggregate	applicable	completed	No exclusions
numeric outliers · fine_aggregate	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0

Check / scope	Applicable	Execution	Reason / exclusions
numeric summary · age_days	applicable	completed	No exclusions
numeric outliers · age_days	applicable	completed	Exclusions: Missing values: 0; Non finite values: 0
regression target · strength_mpa	applicable	completed	No exclusions
regression sample std · strength_mpa	applicable	completed	No exclusions
regression extremes · strength_mpa	applicable	completed	No exclusions
regression bowley skewness · strength_mpa	applicable	completed	No exclusions
regression histogram · strength_mpa	applicable	completed	No exclusions
regression rounding · strength_mpa	applicable	completed	No exclusions
regression constraints · dataset	not applicable	not run	No numerical constraints declared.
regression relationships · dataset	applicable	completed	No exclusions
regression names · dataset	applicable	completed	No exclusions
regression repeated inputs · dataset	applicable	completed	No exclusions
regression availability · dataset	applicable	completed	No exclusions
classification target · dataset	not applicable	not run	Regression uses numerical target evidence and no classification scores.
label conflicts · dataset	not applicable	not run	Regression uses numerical target evidence and no classification scores.
target leakage · dataset	not applicable	not run	Regression uses numerical target evidence and no classification scores.
classification scoring · dataset	not applicable	not run	Regression uses numerical target evidence and no classification scores.
parsing fidelity · dataset	applicable	completed	No exclusions

Coverage describes these checks, not a percentage of universal reliability.

Unsupported: semantic typing; missingness patterns; split/reference comparisons; model performance; non-tabular modalities.

Deterministic fixtures and corpus regression checks provide internal methodology validation. They do not establish external method validity or customer usefulness. Prediction and model-performance analysis belong to ConfidenceGuard.

Resolved source headers

Source position	Original header	Resolved name
1	cement	cement
2	blast_furnace_slag	blast_furnace_slag
3	fly_ash	fly_ash
4	water	water
5	superplasticizer	superplasticizer
6	coarse_aggregate	coarse_aggregate
7	fine_aggregate	fine_aggregate
8	age_days	age_days
9	strength_mpa	strength_mpa

Relationship search ledger

Method / columns	Status / reason	Screened / verified rows
exact copy · cement	verified no relationship	0 / 1030
exact copy · blast_furnace_slag	verified no relationship	0 / 1030
exact copy · fly_ash	verified no relationship	0 / 1030
exact copy · water	verified no relationship	0 / 1030
exact copy · superplasticizer	verified no relationship	0 / 1030
exact copy · coarse_aggregate	verified no relationship	0 / 1030
exact copy · fine_aggregate	verified no relationship	0 / 1030
exact copy · age_days	verified no relationship	0 / 1030
affine identity · cement	verified no relationship	0 / 1030
affine identity · blast_furnace_slag	verified no relationship	0 / 1030
affine identity · fly_ash	verified no relationship	0 / 1030

Method / columns	Status / reason	Screened / verified rows
affine identity · water	verified no relationship	0 / 1030
affine identity · superplasticizer	verified no relationship	0 / 1030
affine identity · coarse_aggregate	verified no relationship	0 / 1030
affine identity · fine_aggregate	verified no relationship	0 / 1030
affine identity · age_days	verified no relationship	0 / 1030
pair sum · cement, blast_furnace_slag	screened no candidate	512 / 0
pair sum · cement, fly_ash	screened no candidate	512 / 0
pair sum · cement, water	screened no candidate	512 / 0
pair sum · cement, superplasticizer	screened no candidate	512 / 0
pair sum · cement, coarse_aggregate	screened no candidate	512 / 0
pair sum · cement, fine_aggregate	screened no candidate	512 / 0
pair sum · cement, age_days	screened no candidate	512 / 0
pair sum · blast_furnace_slag, fly_ash	screened no candidate	512 / 0
pair sum · blast_furnace_slag, water	screened no candidate	512 / 0
pair sum · blast_furnace_slag, superplasticizer	screened no candidate	512 / 0
pair sum · blast_furnace_slag, coarse_aggregate	screened no candidate	512 / 0
pair sum · blast_furnace_slag, fine_aggregate	screened no candidate	512 / 0
pair sum · blast_furnace_slag, age_days	screened no candidate	512 / 0
pair sum · fly_ash, water	screened no candidate	512 / 0
pair sum · fly_ash, superplasticizer	screened no candidate	512 / 0
pair sum · fly_ash, coarse_aggregate	screened no candidate	512 / 0
pair sum · fly_ash, fine_aggregate	screened no candidate	512 / 0
pair sum · fly_ash, age_days	screened no candidate	512 / 0
pair sum · water, superplasticizer	screened no candidate	512 / 0

Method / columns	Status / reason	Screened / verified rows
pair sum · water, coarse_aggregate	screened no candidate	512 / 0
pair sum · water, fine_aggregate	screened no candidate	512 / 0
pair sum · water, age_days	screened no candidate	512 / 0
pair sum · superplasticizer, coarse_aggregate	screened no candidate	512 / 0
pair sum · superplasticizer, fine_aggregate	screened no candidate	512 / 0
pair sum · superplasticizer, age_days	screened no candidate	512 / 0
pair sum · coarse_aggregate, fine_aggregate	screened no candidate	512 / 0
pair sum · coarse_aggregate, age_days	screened no candidate	512 / 0
pair sum · fine_aggregate, age_days	screened no candidate	512 / 0

Histogram count table

Lower	Upper	Count
2.3318078	6.5564087	9
6.5564087	10.78101	42
10.78101	15.005611	66
15.005611	19.230211	68
19.230211	23.454812	65
23.454812	27.679413	103
27.679413	31.904014	85
31.904014	36.128615	106
36.128615	40.353216	115
40.353216	44.577817	95
44.577817	48.802418	52
48.802418	53.027019	56

Lower	Upper	Count
53.027019	57.251619	59
57.251619	61.47622	27
61.47622	65.700821	22
65.700821	69.925422	24
69.925422	74.150023	13
74.150023	78.374624	13
78.374624	82.599225	10

left closed, right open; final bin includes maximum

Grid alignment evidence

Step	Status	Aligned / finite share
1.0	completed	0 / 0
0.1	completed	0 / 0
0.01	completed	0 / 0
0.001	completed	0 / 0

Feature numerical evidence

Feature	Finite summary	IQR evidence
cement · 280 distinct	Finite 1030; min 102; median 272.9; max 540	0
blast_furnace_slag · 187 distinct	Finite 1030; min 0; median 22; max 359.4	2
fly_ash · 163 distinct	Finite 1030; min 0; median 0; max 200.1	0
water · 205 distinct	Finite 1030; min 121.75; median 185; max 247	9
superplasticizer · 155 distinct	Finite 1030; min 0; median 6.35; max 32.2	10
coarse_aggregate · 284 distinct	Finite 1030; min 801; median 968; max 1145	0

Feature	Finite summary	IQR evidence
fine_aggregate · 304 distinct	Finite 1030; min 594; median 779.51; max 992.6	5
age_days · 14 distinct	Finite 1030; min 1; median 28; max 365	59

Evidence references

Evidence ID	Population / denominator	Method / measurements
schema.0	parsed records / 1030	deterministic; /parsing/header_mapping, /profile/row_count, /profile/column_count
missingness.1	all parsed cells / 9270	deterministic; /profile/missing_cells, /profile/missing_rate, /profile/missing_feature_row_rate
column_variation.2	all parsed columns / 9	deterministic; /profile/columns
categorical_cardinality.3	nonnumeric parsed columns / 0	deterministic; /profile/columns
feature_missingness.4	parsed feature cells / 8240	deterministic; /profile/missing_feature_cells, /profile/missing_feature_cell_rate, /profile/missing_feature_row_rate
exact_duplicates.5	all parsed rows / 1030	deterministic; /profile/duplicate_rows, /profile/duplicate_group_count, /profile/rows_in_duplicate_groups Bounded parsed-row references: 73, 78, 81, 84, 87
numeric_summary.6	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/0/numeric_summary, /profile/columns/0/finite_count
numeric_outliers.7	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/0/outlier_count, /profile/columns/0/outlier_rate
numeric_summary.8	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/1/numeric_summary, /profile/columns/1/finite_count
numeric_outliers.9	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/1/outlier_count, /profile/columns/1/outlier_rate
numeric_summary.10	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/2/numeric_summary, /profile/columns/2/finite_count
numeric_outliers.11	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/2/outlier_count, /profile/columns/2/outlier_rate

Evidence ID	Population / denominator	Method / measurements
numeric_summary.12	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/3/numeric_summary, /profile/columns/3/finite_count
numeric_outliers.13	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/3/outlier_count, /profile/columns/3/outlier_rate
numeric_summary.14	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/4/numeric_summary, /profile/columns/4/finite_count
numeric_outliers.15	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/4/outlier_count, /profile/columns/4/outlier_rate
numeric_summary.16	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/5/numeric_summary, /profile/columns/5/finite_count
numeric_outliers.17	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/5/outlier_count, /profile/columns/5/outlier_rate
numeric_summary.18	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/6/numeric_summary, /profile/columns/6/finite_count
numeric_outliers.19	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/6/outlier_count, /profile/columns/6/outlier_rate
numeric_summary.20	finite values; zero/negative/non-finite rates use total rows / 1030	Finite-value descriptive statistics; /profile/columns/7/numeric_summary, /profile/columns/7/finite_count
numeric_outliers.21	finite numeric observations / 1030	Finite-value 1.5-IQR fences; /profile/columns/7/outlier_count, /profile/columns/7/outlier_rate
regression_target.22	finite targets; coverage uses all parsed rows / 1030	finite regression distribution; /profile/regression/target_statistics
regression_sample_std.23	finite targets / 1030	deterministic; /profile/regression/target_statistics/sample_std
regression_extremes.24	finite targets / 1030	deterministic; /profile/regression/target_statistics/extremes Bounded parsed-row references: 1, 154, 182, 382
regression_bowley_skewness.25	finite targets / 1030	deterministic; /profile/regression/target_statistics/bowley_skewness
regression_histogram.26	finite targets / 1030	deterministic; /profile/regression/target_statistics/histogram
regression_rounding.27	finite targets / 1030	deterministic; /profile/regression/target_statistics/rounding

Evidence ID	Population / denominator	Method / measurements
regression_relationships.28	eligible numerical features; explicitly bounded candidate search / not applicable	bounded arithmetic identities; /profile/regression/search
regression_names.29	feature names; contextual cues only / 8	deterministic; /profile/regression/name_candidates
regression_repeated_inputs.30	finite targets grouped by exact feature values / 1030	deterministic; /profile/regression/repeated_inputs
regression_availability.31	user-declared feature context / 8	deterministic; /profile/regression/feature_availability
parsing_fidelity.32	bounded source prefix / 1800	bounded source token comparison; /parsing/fidelity_inspection, /parsing_warnings